List Decoding Reed-Solomon Codes in the Lee, Euclidean, and Other Metrics

ITCS 2026

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Generalized Reed-Solomon Codes

GRS Code: n blocklength, $\mathbb{F}_q$ finite field of size $q \geq n$, k dimension,

$\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_n) \in \mathbb{F}_q^n$ evaluation points, $\boldsymbol{t} = (t_1, \dots, t_n) \in \mathbb{F}_q^n$ non-zero twist factors

$$GRS_{q,k}(\boldsymbol{\alpha}, \boldsymbol{t}) \coloneqq \left\{ \left(t_1 \cdot f(\alpha_1), \dots, t_n \cdot f(\alpha_n) \right) : f \in \mathbb{F}_q[x], \deg(f) < k \right\} \subseteq \mathbb{F}_q^n.$$

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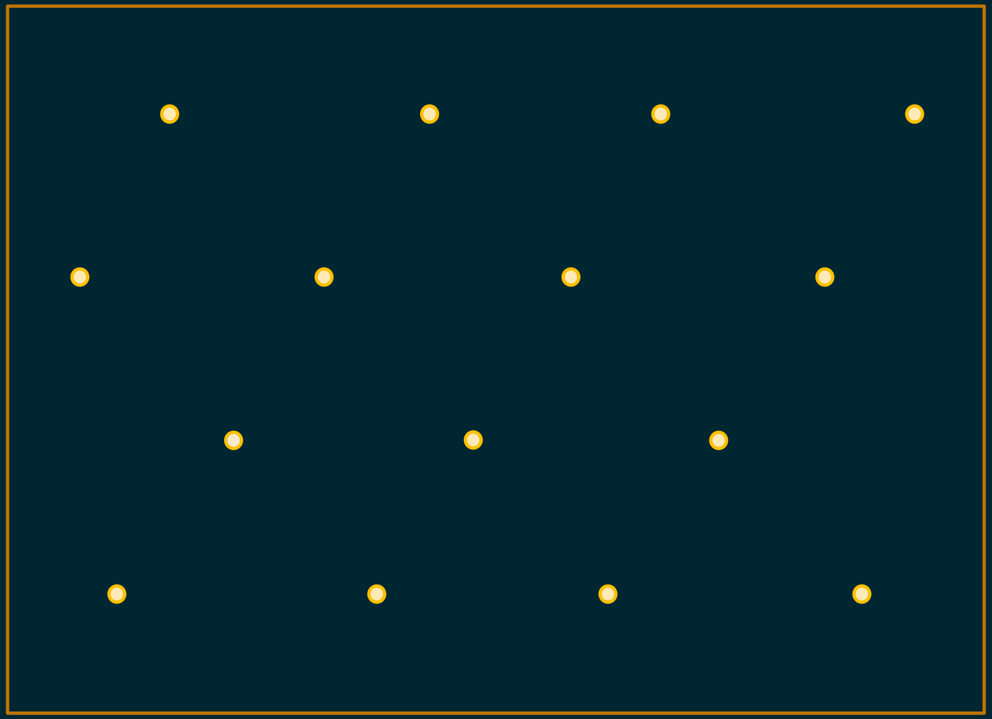
List-Decoding Problem

$\mathcal{C}$ code



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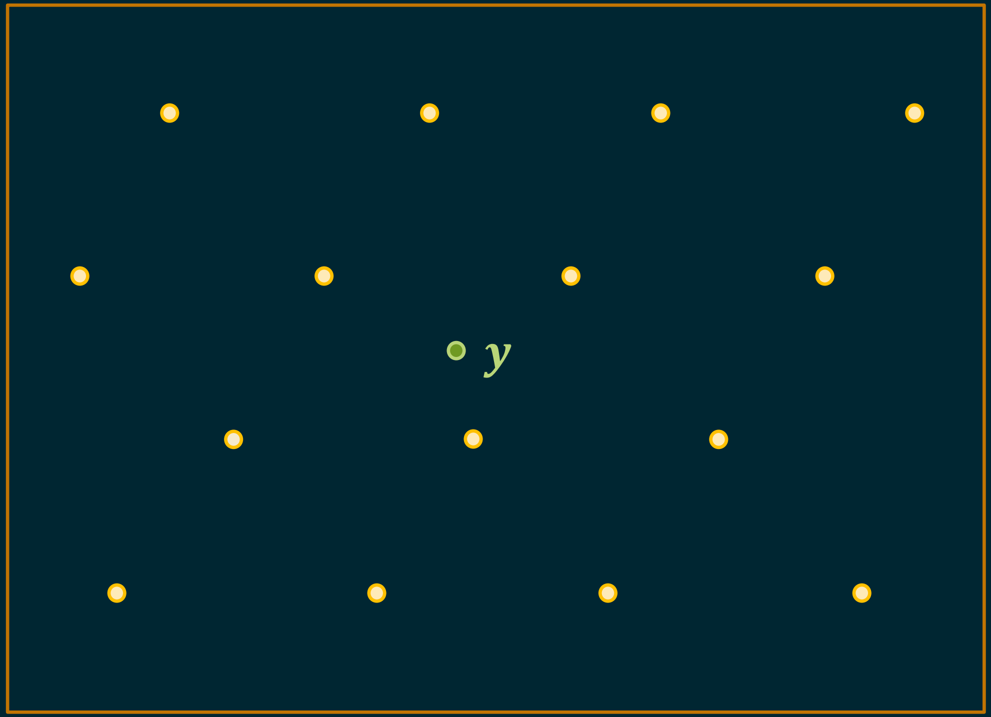

Given a received word y

List-Decoding Problem

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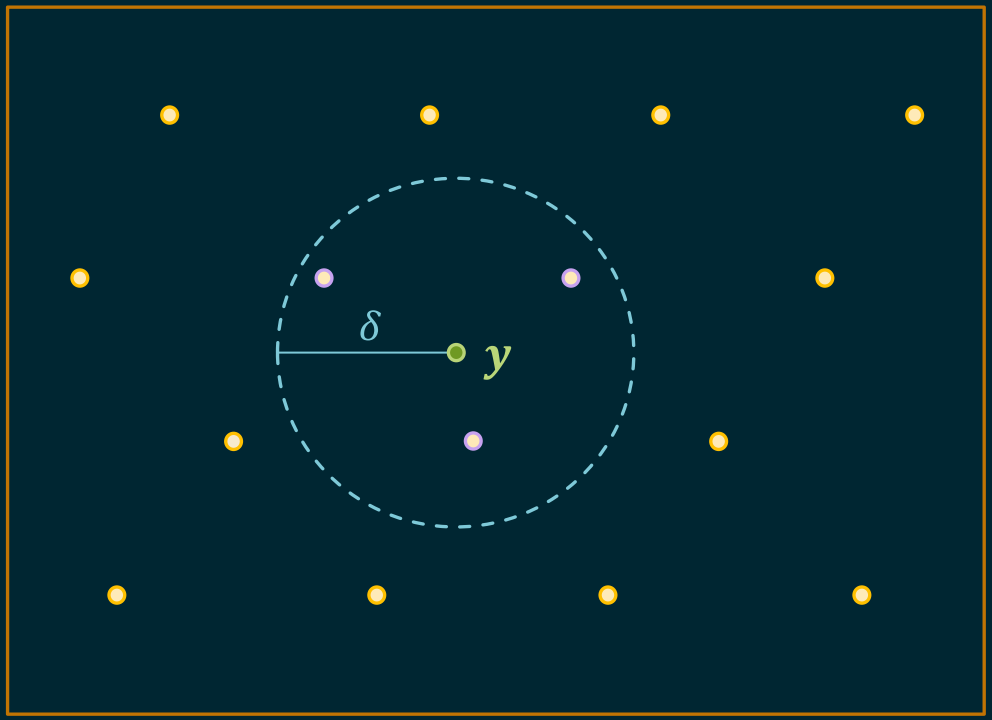
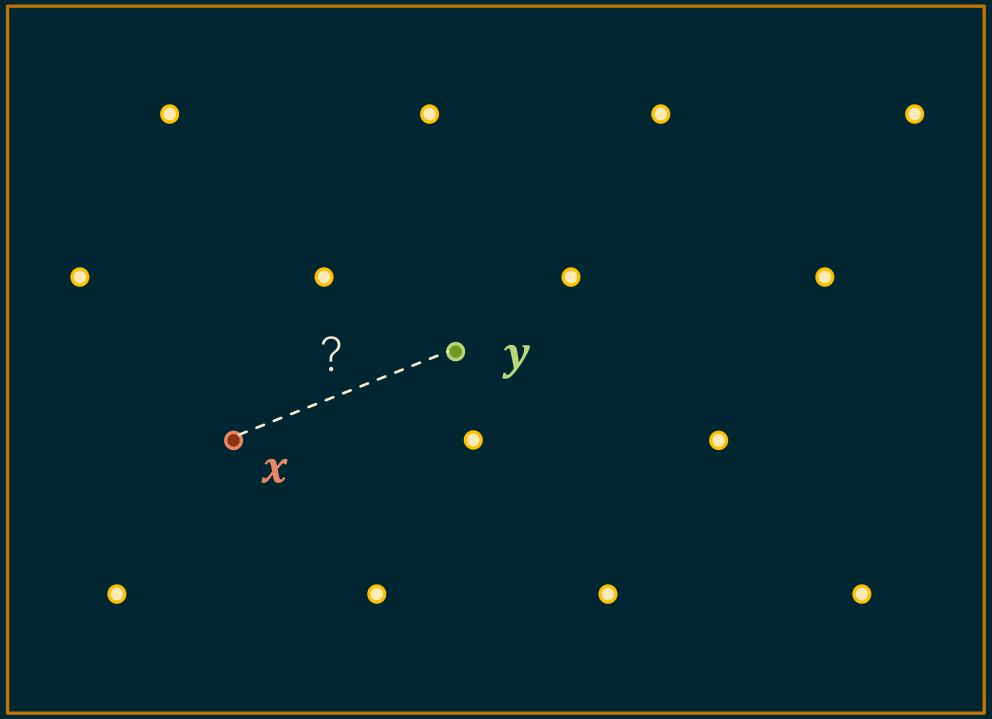


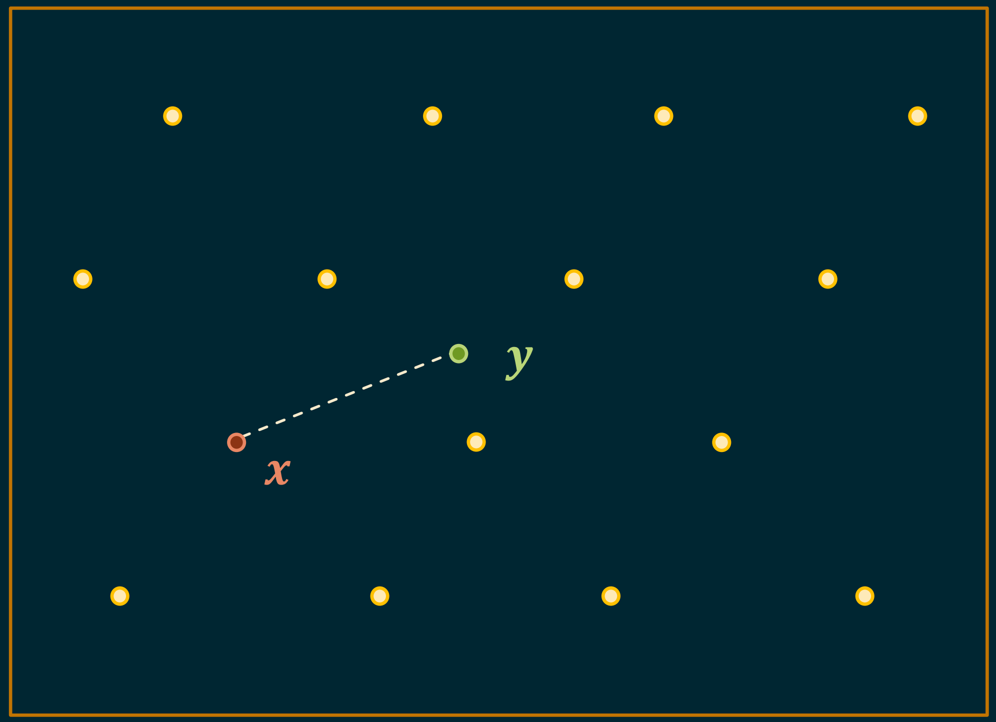
find all codewords within distance δ of $\boldsymbol{y}$.

Measuring Distance



How is distance measured?

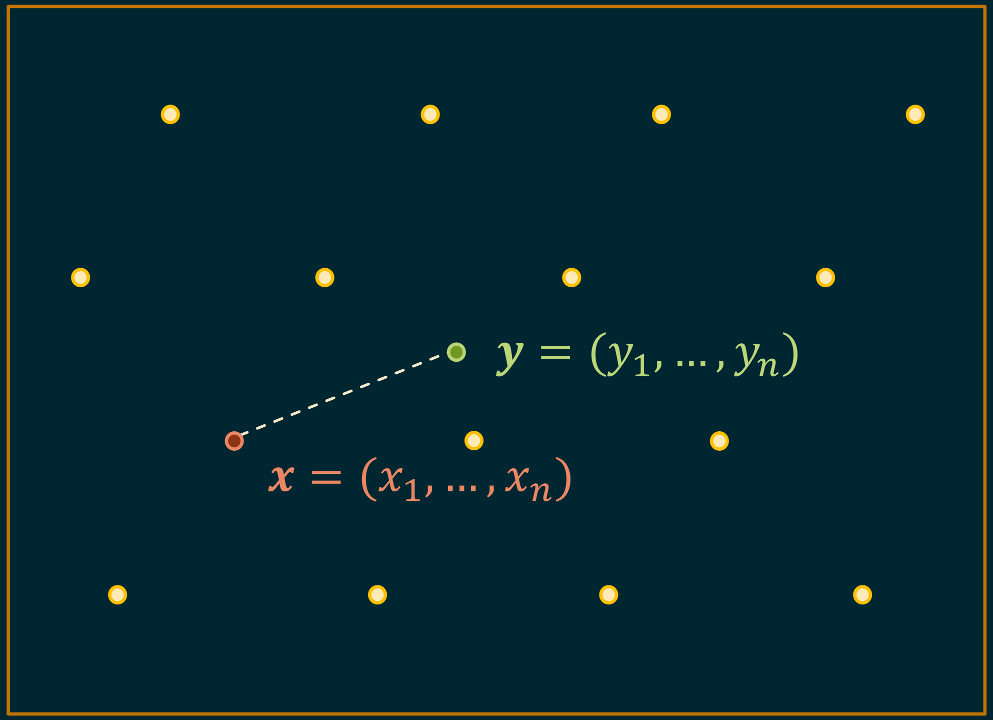
Measuring Distance



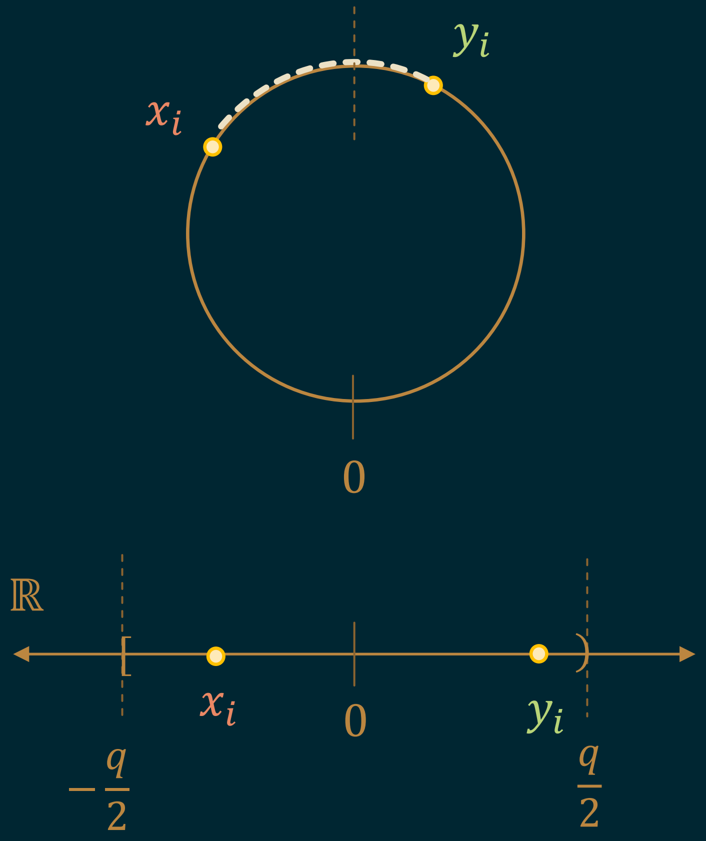
Often in Hamming metric,

but we consider Lee (ℓ_1), Euclidean (ℓ_2), and other ℓ_p metrics.

ℓ_p (Semi)Metric



$\mathbb{R}_q = \mathbb{R}/q\mathbb{Z}$ for q prime



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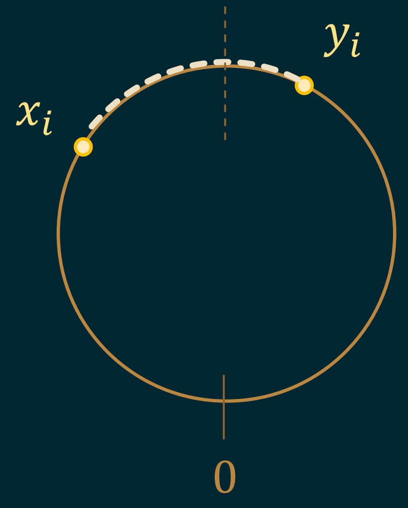

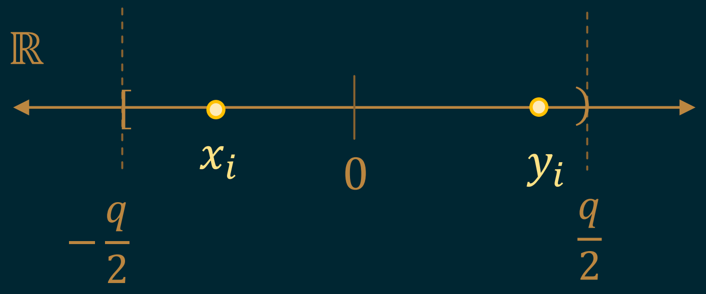

ℓ_p (semi)metric: (on $\mathbb{R}_q^n$) For any $p > 0$ and $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}_q^n$,

$$d(\boldsymbol{x},\boldsymbol{y}) := \underbrace{\|\overline{\boldsymbol{x}-\boldsymbol{y}}\|_p}_{\ell_p\text{ (quasi)norm on }\mathbb{R}^n\text{ given by}}.$$

$$\|\boldsymbol{z}\|_p := \left(z_1^p + \cdots + z_n^p\right)^{1/p}$$

Relative decoding distance: $\delta = d/n^{1/p}$.

Our (Informal) Results

Theorem: *(worst-case)* There is an efficient algorithm that list-decodes any prime-field GRS code from *continuous error* in the Lee, Euclidean, or other ℓ_p (semi)metric for $0 < p \leq 2$, up to *arbitrarily large* (relative) distance $\delta > 0$ for corresponding small enough rate R.

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Theorem: *(average-case)* …and similarly for Laplacian / Gaussian channels, with better rate-distance trade-off than for the Lee / Euclidean metrics (respectively).

Prior Algorithms

Work	Metric	Codes	Decoder Type	Error Type	Rel. Decoding Distance (δ)
[Guruswami-Sudan, 1999]	Hamming	GRS	list	discrete	$\leq 1-\sqrt{R}$
[Mook-Peikert, 2022]	Euclidean (ℓ_2)	prime-field GRS	list	continuous	$\leq \sqrt{(1-R)/2}$
[Roth-Siegel, 1994]	Lee (ℓ_1)	subclass of GRS, BCH	unique	discrete	$\leq 1-R$
[Wu-Kuijper-Udaya, 2003]	Lee (ℓ_1)	prime-field GRS	list	discrete	$\leq g(R)$ piecewise
[Peikert-V. Hostetler, 2025]	any ℓ_p, $0 < p \leq 2$	prime-field GRS	list	continuous	$\leq 1/(R \cdot c_p (e\, p)^{1/p})$

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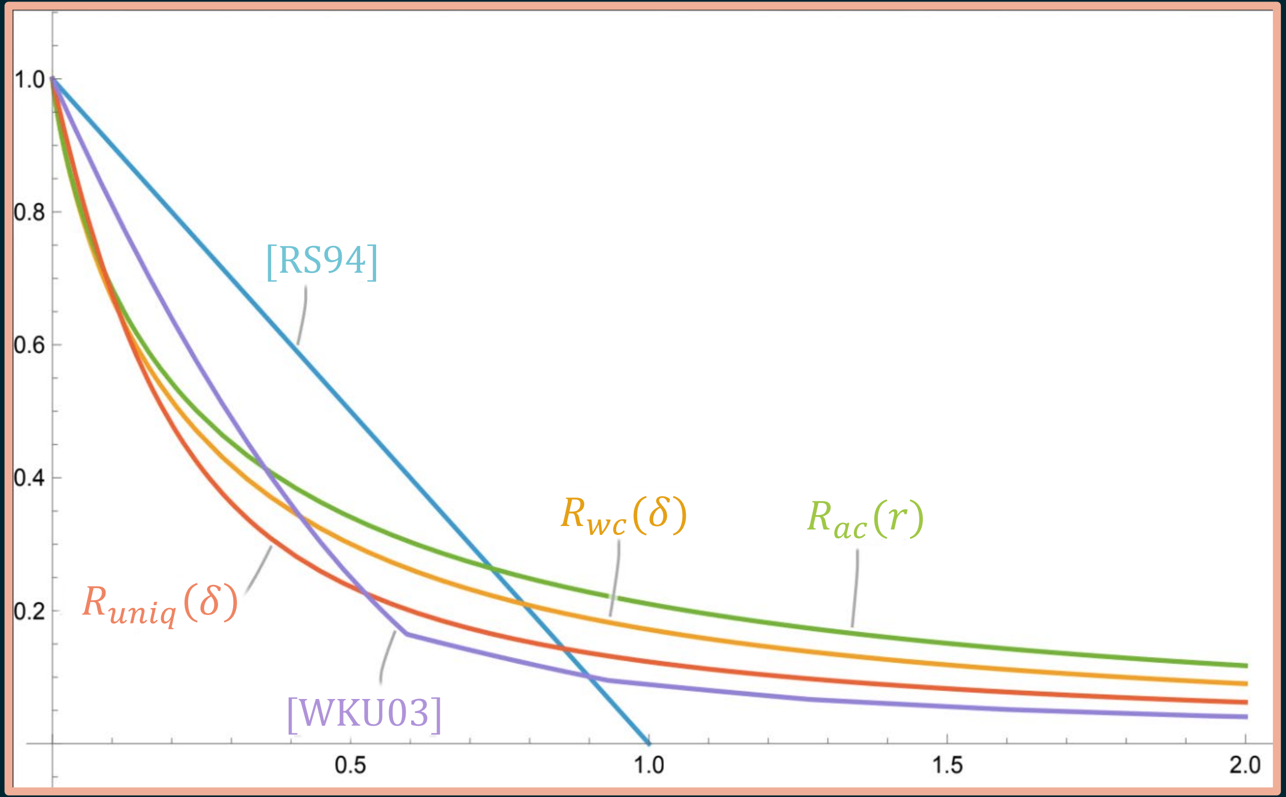
(dimension-normalized) volume of n-dim. ℓ_p ball of radius $n^{1/p}$

Comparison to Prior Algorithms

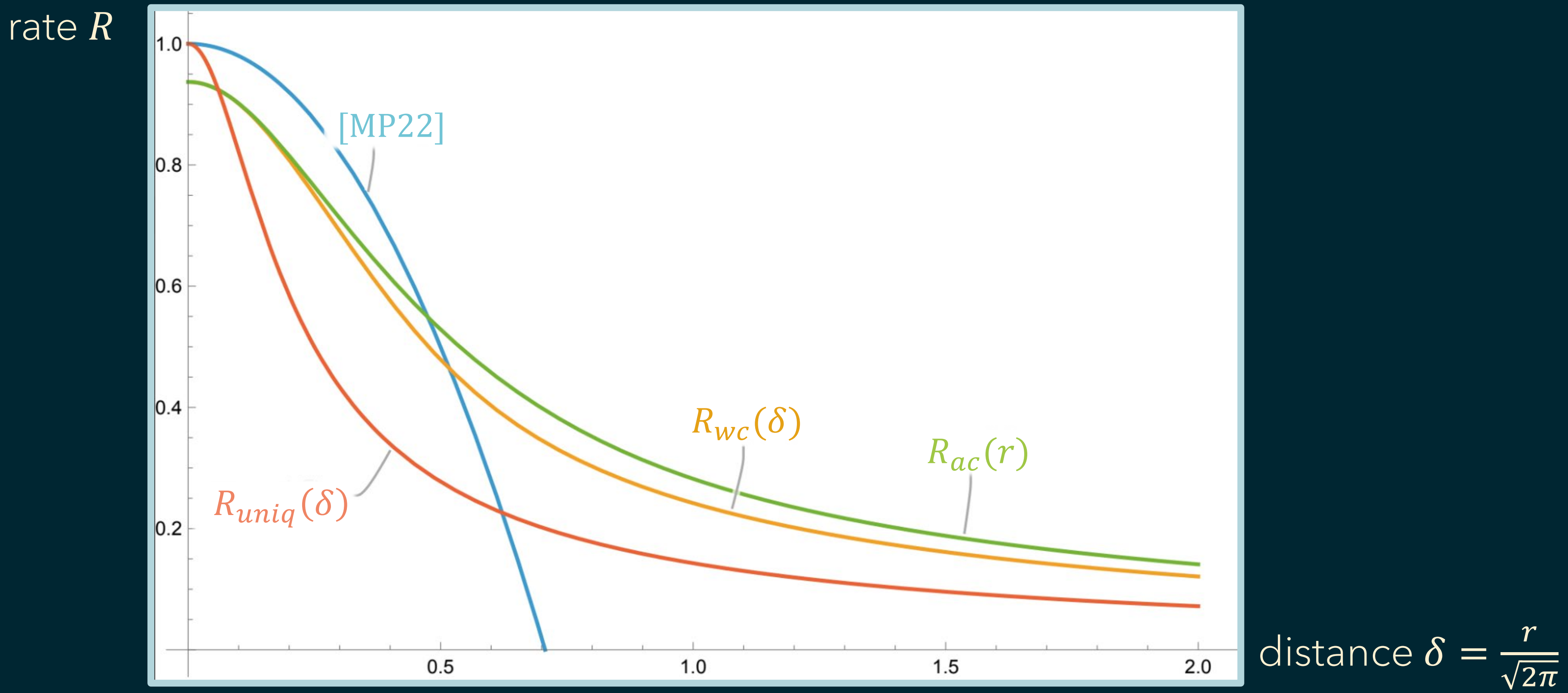
rate R



distance $\delta = \frac{r}{2}$

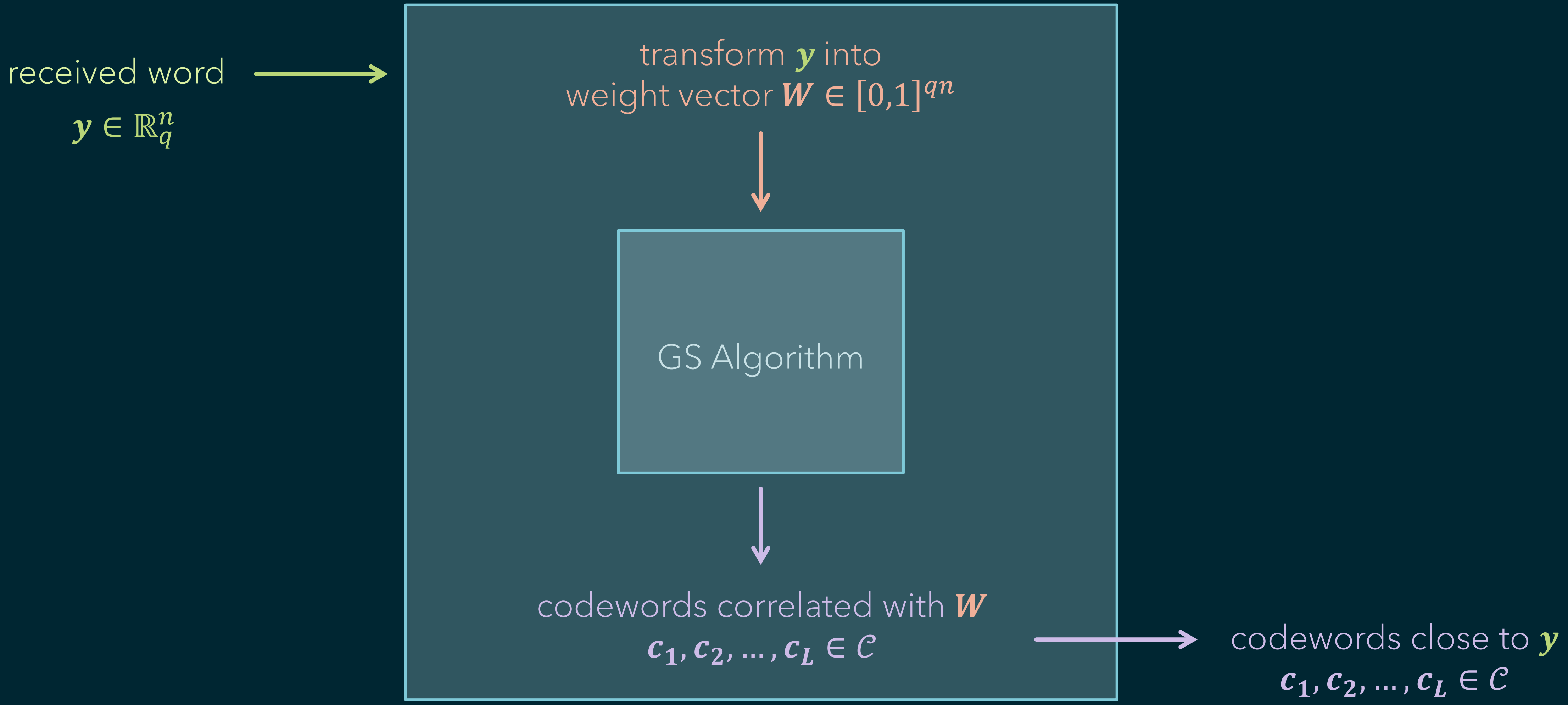
Rate-distance trade-off for Lee (ℓ_1) metric

Comparison to Prior Algorithms



Rate-distance trade-off for Euclidean (ℓ_2) metric

Our List-decoding Algorithm



Transforming into Weights

received word

$\boldsymbol{y} = \boxed{y_1 \mid y_2 \mid \dots \mid y_n} \in \mathbb{R}_q^n$

transform $\boldsymbol{y}$ into
weight vector $\boldsymbol{W} \in [0,1]^{qn}$

GS Algorithm

codewords correlated with $\boldsymbol{W}$
$\boldsymbol{c_1}, \boldsymbol{c_2}, \dots, \boldsymbol{c_L} \in \mathcal{C}$

codewords close to $\boldsymbol{y}$
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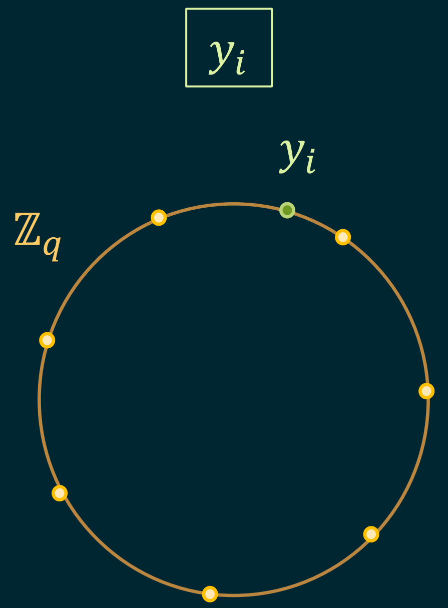

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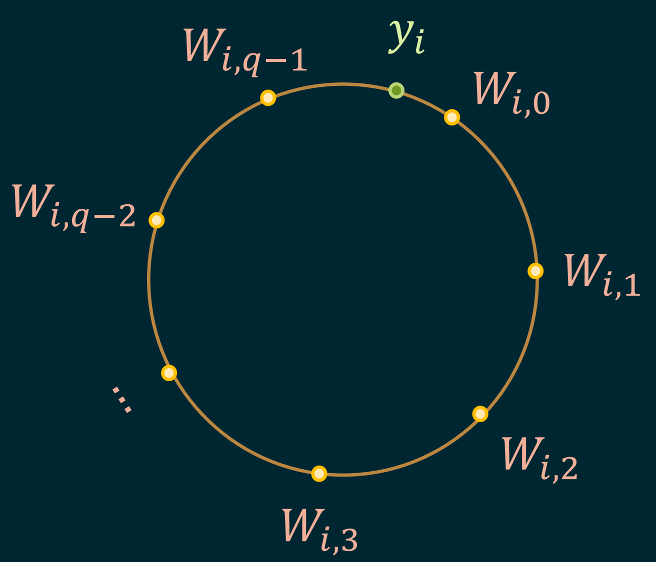
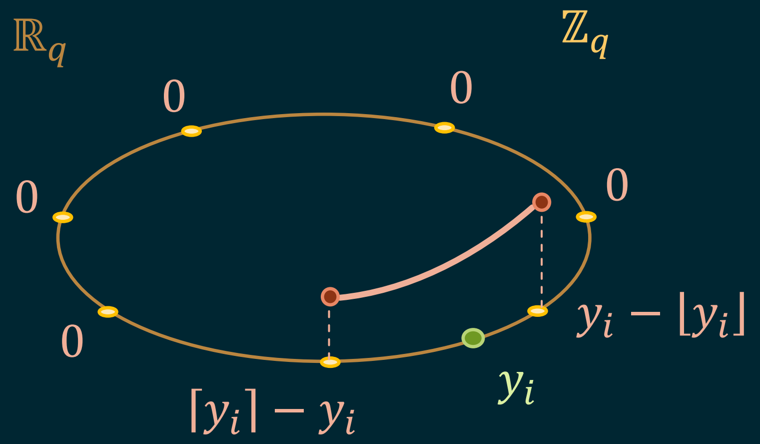
$\boxed{y_i}$



$W_{i,0}$	$W_{i,1}$	$W_{i,2}$	…	$W_{i,q-2}$	$W_{i,q-1}$

Transforming into Weights

[Mook-Peikert, 2022] :



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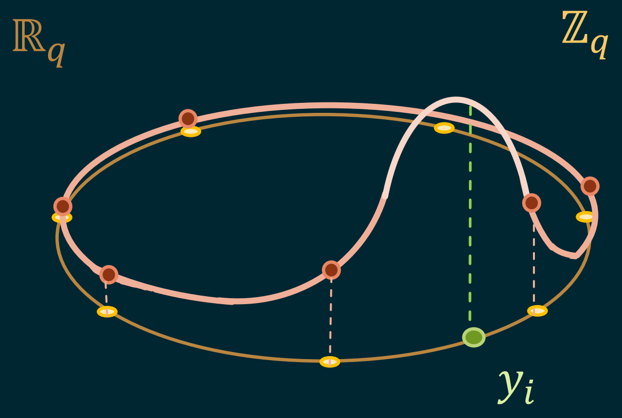
Our weight function :



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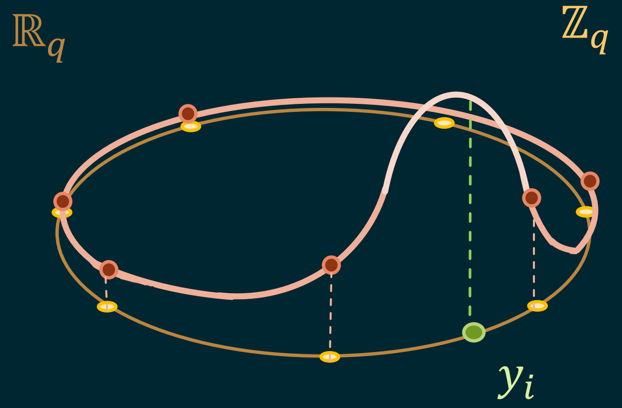


weights given by a

function f_s of width $s > 0$

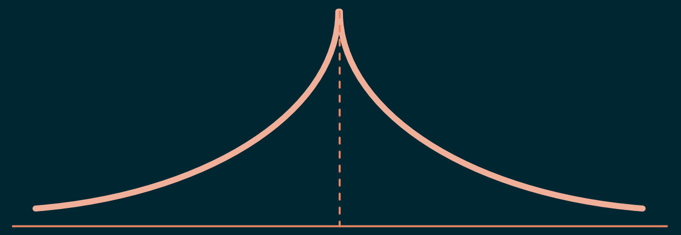
$$f_s(x) \coloneqq \exp(-(c_p|x/s|)^p)$$

Our Weight Function

For distances measured in the Lee (ℓ_1) metric:

$$f_s(x) \coloneqq \exp(-2\,|x/s|^1)$$

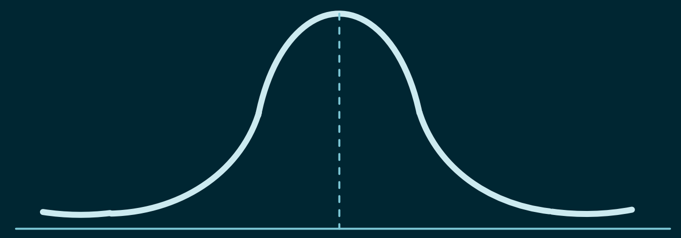
Laplacian function



For distances measured in the Euclidean (ℓ_2) metric:

$$f_s(x) \coloneqq \exp(-\pi\,|x/s|^2)$$

Gaussian function



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We show that any codeword $\boldsymbol{c}$
close enough to $\boldsymbol{y}$, i.e.

$$\text{rel dist}(\boldsymbol{c}, \boldsymbol{y}) \le \delta$$

has large enough correlation
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$$\text{corr}(\boldsymbol{c}, \boldsymbol{W}) > \sqrt{R}.$$

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Then we optimize width s of weight function f_s to maximize R.

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Open Directions

- The product of the rate R and distance δ for which our algorithm works approaches

  $$R \cdot \delta \;\longrightarrow\; 1 \,/\, \text{volume of the } n\text{-dim. } \ell_p \text{ ball of radius } n^{1/p} \text{ (dimension-normalized).}$$

  Why should this be the case?

- What is the list-decoding capacity for decoding over general ℓ_p norms?

  How do our algorithmic bounds compare?

Thank You